CCDC Hunts: Alert Response Playbook

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Every alert in the CCDC Hunts Splunk app has a section here with the same four parts: what it means, how to confirm it, how to scope it, and what to do. Find the alert in the quick reference, click its name, and work it top to bottom. Change the password before you clean up, or the red team logs back in and redoes everything. Open this playbook from Splunk's Playbook menu and every Splunk search in it becomes a link that runs the search. Hold Ctrl when clicking to open it in a new tab.

Alert quick reference

Click an alert's name to jump to its steps. Names match the alerts in Splunk exactly. Severity runs from 3 (low) to 6 (critical).

Windows remote execution

Alert	Severity	Usually means
smbexec service runs cmd	6 Critical	Admin password used to run commands
smbexec cmd started by services.exe	5 High	smbexec's command actually ran
Shell under WMI, WinRM or Task Scheduler	4 Medium	wmiexec, WinRM shell or atexec
Batch file written to Windows Temp	4 Medium	smbexec's temporary .bat
whoami run	4 Medium	First command after getting a shell

Windows accounts and logons

Alert	Severity	Usually means
DCSync by a non-DC account	6 Critical	Every password hash dumped
Account added to an admin group	5 High	Keeping admin after password changes

Alert	Severity	Usually means
Account locked out	5 High	Guessing, or knocking out a scored account
New Windows account created	4 Medium	Backdoor account
Password spraying	4 Medium	Guessing passwords across accounts
Kerberoasting	4 Medium	Cracking a service account offline

Windows persistence and tampering

Alert	Severity	Usually means
Event log cleared	6 Critical	Covering their tracks
Defender turned off or exclusion added	5 High	About to drop tools
Defender detected malware	5 High	Tools already on the box
Autorun or IFEO registry change	5 High	Runs at logon or boot, or sticky keys
WMI event subscription created	5 High	Hidden way back in
Logging stopped or reconfigured	5 High	Blinding you first
Script host running a disguised file	5 High	JScript hidden in log-named files
Scheduled task created	4 Medium	Payload on a timer
Suspicious PowerShell	4 Medium	Download or decode code

Linux

Alert	Severity	Usually means
Linux user or group changed	5 High	Backdoor user or new sudo member
Root logged in over SSH	5 High	Stolen root password or planted key
SSH brute force	4 Medium	Guessing SSH passwords

Alert	Severity	Usually means
Crontab edited	4 Medium	Backdoor restarting on a timer

Web, email and FTP

Alert	Severity	Usually means
Website returning server errors	5 High	Scored site broken
Web attack patterns	4 Medium	Scanning or exploiting the site
Mail login failures	3 Low	Guessing mail passwords
FTP login failures	3 Low	Guessing FTP passwords

Network

Alert	Severity	Usually means
Firewall or router config changed	5 High	Rules or passwords changed
Firewall threat detected	4 Medium	Exploit, malware or C2 traffic

Health and Splunk

Alert	Severity	Usually means
Host stopped sending logs	5 High	You're blind on that host
Splunk login failures	4 Medium	Someone going after Splunk

How to work any alert

Every alert gets the same six steps, in order. Containment means passwords, not IP blocks: the red team often shares the scoring engine's IP.

1. **Confirm.** Rule out the scoring engine and your teammates. Ask in team chat who was doing what at that minute.
2. **Scope.** Find the host, account, source IP and what they ran (see Splunk pivots).

3. **Tell the box owner** in one line: `11:42 · DC · smbexec ran "whoami" · Administrator · from 10.0.0.66`
4. **Contain.** Reset the account's password, then close its sessions.
5. **Clean up.** Disable and quarantine instead of deleting, then run the persistence sweep.
6. **Verify and write it up.** Re-test the old password from Kali (it should fail), watch the dashboard for 30 minutes, fill in the incident report.

Quarantine a file on Windows (hash it first):

```
New-Item -ItemType Directory C:\ccdc\quarantine -Force | Out-Null
Get-FileHash "<file>" | Format-List
Move-Item "<file>" C:\ccdc\quarantine\
```

On Linux: `sudo mkdir -p /root/quarantine && sha256sum <file> && sudo mv <file> /root/quarantine/`

Windows remote execution

Someone is running commands on a Windows box from another machine. Dashboard: Windows → Remote Execution (`/app/ccdc_hunts/remote_exec`).

smbexec service runs cmd

Severity 6 · System log 7045

Someone with an admin password created a service that runs cmd, a .bat file or PowerShell. smbexec makes one for every command, so treat this as an intruder who is active right now.

Confirm. Read `Service_File_Name` . The text right after `echo` is their actual command. It's only a false alarm if a teammate installed software on that box at that minute.

Scope. Find the account and IP that logged in at the same minute:

Run in Splunk »

```
`windows_security` host="<host>" EventCode=4624 Logon_Type=3 | eval
user=mvindex(Account_Name,-1) | search NOT user="*$" | table _time user
Source_Network_Address
```

Respond.

1. Reset that account's password. Close its SMB sessions: `Get-SmbSession | Where-Object ClientComputerName -eq '<IP>' | Close-SmbSession -Force`

2. Disable any leftover service: `Get-CimInstance Win32_Service | Where-Object PathName -match 'COMSPEC|cmd\.exe|\.bat' | Select-Object Name, PathName`, then `Set-Service <name> -StartupType Disabled`
3. Quarantine the six-letter output files at the root of C: (`Get-ChildItem C:\ -File | Where-Object Name -match '^[A-Za-z]{6}$'`).
4. Undo whatever their commands did, run the persistence sweep, and confirm `nxc smb <host IP> -u <user> -p '<old password>'` now fails.

smbexec cmd started by services.exe

Severity 5 · Sysmon 1

cmd or PowerShell was started directly by `services.exe`. This is smbexec's service actually running, so it usually fires together with the alert above.

Confirm. `CommandLine` looks like `cmd.exe /Q /c %TEMP%\XyZaBc.bat`. A few old services start through a cmd wrapper, but they repeat the same command every time. smbexec's names are random.

Scope. See what the shells ran next:

Run in Splunk »

```
`windows_sysmon` host="<host>" EventCode=1 ParentImage="*cmd.exe" | table
_time User ParentCommandLine Image CommandLine
```

Respond. Same as smbexec service runs cmd. Quarantine anything the scope search shows them starting.

Shell under WMI, WinRM or Task Scheduler

Severity 4 · Sysmon 1

A shell or script host was started through WMI, WinRM or Task Scheduler. The `ParentImage` column tells you which:

Parent	Usually	Respond
`WmiPrvSE.exe`	wmiexec	Reset the account's password; keep port 135 blocked at the firewall

Parent	Usually	Respond
`wsmprovhost.exe`	WinRM (evil-winrm, nxc winrm) or the scoring check	Compare with the scorer's normal command below; reset the password; lock WinRM to named accounts
`taskeng.exe` or `svchost.exe -s Schedule`	atexec, or a planted task	Export the task, then disable it

- [ ] Scorer's normal WinRM command (write it here so nobody chases it):

Scope. Who used WinRM recently: `Get-WinEvent -LogName Microsoft-Windows-WinRM/Operational -MaxEvents 200 | Where-Object Id -eq 169 | Select-Object TimeCreated, Message`

Respond to a planted task (undo with `Enable-ScheduledTask`):

```
Export-ScheduledTask -TaskName '<name>' -TaskPath '<path>' >
C:\ccdc\task_<name>.xml
Disable-ScheduledTask -TaskName '<name>' -TaskPath '<path>'
```

Batch file written to Windows Temp

Severity 4 · Sysmon 11

A `.bat` file appeared in `C:\Windows\Temp`. smbexec writes one per command and deletes it a second later.

Confirm. `cmd.exe` writing a six-random-letter `.bat` (like `CDfFIE.bat`) is smbexec. Installers sometimes drop `.bat` files with readable names.

Respond. Work it as smbexec service runs cmd. If this fires on its own, check the host's System log is still reaching Splunk on the Health dashboard.

whoami run

Severity 4 · Sysmon 1 and Security 4688

`whoami` is usually the first thing an attacker runs after getting a shell. Each run can show twice, once from each log. The Security log is the backup if Sysmon gets killed.

Confirm by reading the `parent` column:

Parent	What it means
`cmd.exe` under `services.exe`	smbexec: work smbexec service runs cmd
`cmd.exe` under `WmiPrvSE.exe`	wmiexec: work Shell under WMI, WinRM or Task Scheduler
`wsmprovhost.exe`	WinRM: the red team or the scoring check
`explorer.exe` or a console window	Someone at the keyboard or on RDP; ask your team
PowerShell or a script host with an odd parent	A payload; find what started it

Respond. Follow the matching alert. If it came from a payload, quarantine the parent's file, check its connections (Sysmon 3), then run the persistence sweep.

Windows accounts and logons

Someone is stealing, guessing or abusing account access. Dashboard: Windows → Accounts & Logons (`/app/ccdc_hunts/accounts`).

DCSync by a non-DC account

Severity 6 · Security 4662, on the DC

A normal account asked the DC to hand over password data the way another DC would. This is how secretsdump and mimikatz dump every password hash in the domain. Assume every password is known.

Confirm. The `user` column is a normal account, not a computer account ending in `$`. This alert needs Directory Service Access auditing on the DC. Test it in the lab by running `impacket-secretsdump` against your DC and checking it fires.

Respond.

1. Reset the account it used and remove that account from every admin group.
2. Rotate every admin and service account password. Change scoring accounts only through the competition's process.

3. Ask the captain about resetting the `krbtgt` password twice (Microsoft's New-KrbtgtKeys.ps1 script). It kills forged Kerberos tickets, but briefly breaks Kerberos for everyone, so watch the scoreboard after.

Account added to an admin group

Severity 5 · Security 4728, 4732, 4756

An account was added to Domain Admins, Administrators or another powerful group. Red teams do this so they keep admin rights after you change passwords.

Confirm. Check `changed_by`. If a teammate did it on purpose, write it down. Otherwise it's the red team.

Respond.

1. Remove the member: `Remove-ADGroupMember "<group>" -Members <user>` (a local group: `Remove-LocalGroupMember -Group Administrators -Member <user>`).
2. Reset the password of the `changed_by` account. They are using it.
3. If the added account is new, disable it: `Disable-ADAccount <user>` (undo: `Enable-ADAccount`).

Account locked out

Severity 5 · Security 4740, on the DC

An account hit the lockout limit. If it's a scoring account, that service goes red, and red teams lock scoring accounts on purpose.

Confirm. `Caller_Computer_Name` shows where the bad passwords came from. Check whether `locked_account` is one the scoring engine uses.

Respond.

1. Unlock it: `Unlock-ADAccount <user>`
2. If it keeps happening, find the source on the Failed logons panel.
3. Talk to the captain before changing the lockout policy (`Get-ADDefaultDomainPasswordPolicy` shows it). A threshold of 0 stops lockouts, but makes password guessing free.

New Windows account created

Severity 4 · Security 4720

Someone created an account. Red teams make backdoor accounts with ordinary names like `svc_backup` or `helpdesk2`.

Confirm. Check `created_by`. A teammate, or an inject asking for new users, explains it.

Respond. Disable instead of deleting: `Disable-ADAccount <user>` (local: `Disable-LocalUser <user>`). Make sure it isn't in any admin group. If your team didn't create it, reset the `created_by` account's password.

Password spraying

Severity 4 · Security 4625

One source IP failed 20 or more logons, or tried 5 or more usernames, in 5 minutes. They're guessing passwords.

Confirm. `tried` lists the usernames. The scorer only fails logons if you changed its password without updating the team portal.

Respond. Make sure no account in `tried` still has its default password. Watch for a successful logon (4624) from the same IP on the Remote logons panel. Only block the IP if you're sure it isn't the scorer's.

Kerberoasting

Severity 4 · Security 4769, on the DC

Someone requested a Kerberos ticket with weak RC4 encryption for a service account. Attackers do this to crack the service account's password offline.

Confirm. `Service_Name` is a user account that runs a service, often SQL or IIS. Many requests from one `Client_Address` in a short time is not normal.

Respond.

1. Give the targeted account a long random password (25+ characters) so cracking fails.
2. If a service runs as that account, update the password in the service's settings too (services.msc, or SQL Server Configuration Manager for MSSQL), then check its scored service is still green.
3. Look at everything else `Client_Address` did with the Splunk pivots.

Windows persistence and tampering

Someone is setting up a way back in, or trying to blind you. Dashboard: Windows → Persistence & Tampering (`/app/ccdc_hunts/persistence`).

Event log cleared

Severity 6 · Security 1102, System 104

Someone wiped a Windows event log to hide what they did. Nothing legitimate does this during a competition.

Confirm. `who` shows the account. Splunk still has everything that was forwarded before the wipe.

Respond. Reset that account's password. Use the Splunk pivots on the host for the 30 minutes before the clear, because that's what they were hiding. Then run the persistence sweep.

Defender turned off or exclusion added

Severity 5 · Defender 5001, 5010, 5012, 5007

Real-time protection was turned off, or a folder or program was excluded from scanning. Attackers do this right before dropping tools.

Confirm. For 5007, `Message` shows the old and new value. An exclusion appears under `Exclusions` .

Respond. Undo it, then scan the excluded path, because that's where their tools are:

```
Get-MpPreference | Select-Object ExclusionPath, ExclusionProcess,
ExclusionExtension, DisableRealtimeMonitoring
Remove-MpPreference -ExclusionPath '<path>'
Set-MpPreference -DisableRealtimeMonitoring $false
Start-MpScan -ScanType CustomScan -ScanPath '<path>'
```

Defender detected malware

Severity 5 · Defender 1116, 1117

Defender found malware (1116) or acted on it (1117).

Confirm. `Path` is the file. `Process_Name` is what ran or wrote it.

Scope. Find who dropped the file:

Run in Splunk »

```
`windows_sysmon` host="<host>" EventCode=11 TargetFilename="*<file name>*" |
table _time User Image TargetFilename
```

Respond. Check that Defender quarantined it (a 1117 event). Treat the host as compromised: reset the account that dropped it and run the persistence sweep.

Autorun or IFEO registry change

Severity 5 · Sysmon 12, 13, 14

Someone set a registry value that runs a program at logon or boot (Run keys, Winlogon), or the sticky-keys backdoor (an Image File Execution Options `Debugger`).

Confirm. `TargetObject` ends with the value's name; the key is everything before the last backslash. `Details` is the program it now runs. Real software sometimes adds Run keys pointing at a readable vendor path.

Respond. Back up the key, delete the value, and quarantine the program from `Details`:

```
reg export "<key>" C:\ccdc\regbackup.reg
reg delete "<key>" /v "<value name>" /f
```

Undo with `reg import C:\ccdc\regbackup.reg`.

WMI event subscription created

Severity 5 · Sysmon 19, 20, 21

Someone created a WMI subscription that runs a command when an event happens. It survives reboots and password changes, and nothing in a CCDC image normally creates one.

Confirm. `Destination` or `Consumer` shows the command it runs.

Respond. List them, write down the command for the report, then remove the consumer, filter and binding:

```
Get-CimInstance -Namespace root\subscription -ClassName __EventFilter |
Select-Object Name, Query
Get-CimInstance -Namespace root\subscription -ClassName __EventConsumer |
Select-Object Name, CommandLineTemplate, ScriptText
Get-CimInstance -Namespace root\subscription -ClassName
__FilterToConsumerBinding | Where-Object { $_.Consumer -match '<name>' } |
Remove-CimInstance
Get-CimInstance -Namespace root\subscription -ClassName __EventConsumer -
Filter "Name='<name>'" | Remove-CimInstance
Get-CimInstance -Namespace root\subscription -ClassName __EventFilter -Filter
"Name='<filter name>'" | Remove-CimInstance
```

Logging stopped or reconfigured

Severity 5 · Sysmon 4, 16; System 7036

Sysmon or the Splunk forwarder stopped, or Sysmon's config changed. Red teams blind you before doing anything loud.

Confirm. Sysmon "Started" after a reboot is normal. "Stopped", or a config change (16) your team didn't make, is not.

Respond. Start both again, reload your Sysmon config if it changed, then hunt the time the host was quiet:

```
Get-Service Sysmon64, SplunkForwarder | Select-Object Name, Status, StartType
Start-Service Sysmon64, SplunkForwarder
Sysmon64.exe -c sysmonconfig-export.xml
```

If logging stops completely, you may only get the Host stopped sending logs alert.

Script host running a disguised file

Severity 5 · Sysmon 1

`wscript` or `cscript` was told to run a file as script with `//E:`. This is the invitational red team's payload trick: JScript hidden in files named like logs (`etl_data.log`, `cache_entry.dat`).

Confirm. `//E:JScript` or `//E:VBScript` pointed at a non-script file is malicious.

Respond.

1. Kill it: `Stop-Process -Id <ProcessId>`

2. Quarantine the file named in `CommandLine`.
3. Find what launched it (`ParentImage`, usually a scheduled task), disable that, then run the persistence sweep.

Scheduled task created

Severity 4 · Security 4698

A new scheduled task was registered. Attackers use tasks to run payloads on a timer or when the machine is idle, like the fake `WinSAT_` task from the invitational.

Confirm. Updaters create tasks with vendor names. Random suffixes, names copying real Windows tasks, or actions that run cmd, PowerShell or wscript are bad.

Respond. Look at what it runs, then export and disable it (undo with `Enable-ScheduledTask`):

```
Get-ScheduledTask -TaskName '<name>' | Select-Object TaskPath, @{n='Runs';e=
{$_.Actions.Execute + ' ' + $_.Actions.Arguments}}
Export-ScheduledTask -TaskName '<name>' -TaskPath '<path>' >
C:\ccdc\task_<name>.xml
Disable-ScheduledTask -TaskName '<name>' -TaskPath '<path>'
```

Suspicious PowerShell

Severity 4 · PowerShell 4104

A PowerShell script contained download, decode or Defender-tampering code (`DownloadString`, `IEX`, `FromBase64String`, `Add-MpPreference`, and similar).

Confirm. `Message` holds the actual script. Your team's hardening scripts can trip this, so check who ran it.

Scope. See where PowerShell connected:

Run in Splunk »

```
`windows_sysmon` host="<host>" EventCode=3 Image="*powershell.exe" | table
_time User DestinationIp DestinationPort
```

Respond. Quarantine anything it downloaded, undo any Defender change it made, and reset the account that ran it.

Linux

The same attacks on Rocky, both Ubuntus and the Debian Splunk box. Dashboard: Linux (`/app/ccdc_hunts/linux`).

Linux user or group changed

Severity 5 · auth.log / secure

An account or group was created, deleted, or had members added (`useradd` , `usermod` , `gpasswd`). Red teams add backdoor users or put themselves in `sudo` or `wheel` .

Confirm. The `program` column and the raw line show what changed. A teammate adding users for an inject explains it.

Respond. Check who has admin, remove the member, and lock the account instead of deleting it:

```
getent group sudo wheel
sudo gpasswd -d <user> sudo           # wheel on Rocky
sudo usermod -L -e 1 <user>           # undo: sudo usermod -U -e "" <user>
sudo cat /home/<user>/.ssh/authorized_keys
```

Root logged in over SSH

Severity 5 · auth.log / secure

Someone logged in as root over SSH. Few setups need this. It's what a red team does with a stolen root password or a planted SSH key.

Confirm. `method` shows `password` or `publickey` . A key login you didn't set up means a planted key.

Respond.

1. Change root's password: `sudo passwd root`
2. Back up `/root/.ssh/authorized_keys` , then remove keys you don't recognize.
3. Kick their session: `who` shows the terminal, then `sudo pkill -KILL -t pts/<n>`
4. If the scored SSH check doesn't use root, set `PermitRootLogin no` in `/etc/ssh/sshd_config` and restart SSH (`sshd` on Rocky, `ssh` on Ubuntu and Debian).

SSH brute force

Severity 4 · auth.log / secure

One IP failed 10 or more SSH logins in 5 minutes.

Confirm. `tried` shows the usernames. The scorer only fails if its password changed.

Respond. Make sure none of those accounts has a default or weak password. Watch the SSH logins panel for an `Accepted` login from the same IP. Only block the IP if you're sure it isn't the scorer's.

Crontab edited

Severity 4 · cron log / syslog

Someone edited a user's crontab. Red teams use cron to restart a backdoor every minute.

Confirm. The raw line shows whose crontab, for example `(root) REPLACE (root)`. A teammate editing cron explains it.

Respond. Back it up, then remove the bad line. System cron files aren't logged, so check them too:

```
sudo crontab -l -u <user> | sudo tee /root/cron_<user>.bak
sudo crontab -e -u <user>
ls -la /etc/cron.d /etc/cron.hourly; cat /etc/crontab
```

Web, email and FTP

The scored apps are being attacked or are broken. Dashboard: Web, Email & FTP (`/app/ccdc_hunts/web_mail_ftp`).

Website returning server errors

Severity 5 · web access log

The site returned 10 or more server errors (status 500 or higher) in 5 minutes. The HTTP and HTTPS checks compare the page to an expected copy, so a broken page loses points.

Confirm. `pages` shows which URLs fail. Load the site yourself and check the scoreboard.

Respond. Read the error log before restarting anything:

```
sudo systemctl status httpd nginx php-fpm mariadb     # apache2 on Ubuntu
sudo tail -50 /var/log/httpd/error_log                 # or
/var/log/nginx/error.log
sudo diff -r /var/www/html /root/www_backup            # changed site files
```

The usual causes are a stopped database, a config change by your team, or site files the red team edited or deleted. This is why the web root gets backed up in the first 15 minutes.

Web attack patterns

Severity 4 · web access log

Requests that look like path traversal, SQL injection, command injection or script injection, or that come from a scanning tool (sqlmap, nikto, gobuster).

Confirm. `sample_requests` shows what they tried. A 200 status on an attack request means it may have worked. 404s are just scanning.

Scope. Everything that IP got a 200 for:

Run in Splunk »

```
`web_access` clientip="<IP>" status=200 | table _time method uri status bytes
```

Respond.

- Only 404s: note it, nothing to clean up.
- A 200 on an attack request: list site files changed in the last hour with `sudo find /var/www -type f -mmin -60`. A new `.php` file there is probably a web shell, so quarantine it.
- Check the Possible web shells panel either way.

Mail login failures

Severity 3 · mail log

One IP failed 10 or more mail logins (SMTP, IMAP or POP3) in 5 minutes. Mail uses AD usernames, so this can also lock AD accounts.

Confirm. The scorer only fails if a mail user's password changed.

Respond. Same as Password spraying. Watch the Mail logins panel for a success from the same IP. If outbound mail spikes, the server may be relaying spam: `postqueue -p | tail -1` shows the queue size.

FTP login failures

Severity 3 · vsftpd log

One IP failed 10 or more FTP logins in 5 minutes.

Confirm. Check `anonymous_enable` in `/etc/vsftpd.conf` (`/etc/vsftpd/vsftpd.conf` on Rocky). If anonymous login is on, they don't need to guess.

Respond. Make sure FTP users don't have default passwords. Check the FTP activity panel for uploads: a script uploaded into a folder the website serves is a web shell.

Network

The Palo Alto and the VyOS router. These need both devices sending syslog to Splunk, plus the Palo Alto add-on. Dashboard: Network (`/app/ccdc_hunts/network`).

Firewall or router config changed

Severity 5 · Palo Alto config and system logs, VyOS syslog

Someone committed a config change on the Palo Alto or the VyOS router. Red teams open ports, add NAT rules or change the admin password.

Confirm. Ask your firewall person whether they just committed. `admin` and `cmd` (Palo Alto) or the raw line (VyOS) show who and what.

Respond.

- **Palo Alto:** Monitor → Logs → Configuration shows each change. Device → Config Audit compares two versions. To go back: Device → Setup → Operations → Load configuration version, then Commit. Change the admin password.
- **VyOS:** `show system commit` lists commits and `compare <n>` shows what changed. Fix it in `configure` mode with `delete` or `set`, then `commit` and `save`. Change the admin password.

Firewall threat detected

Severity 4 · Palo Alto threat log

The Palo Alto matched a high or critical threat signature: an exploit, malware, or command-and-control traffic.

Confirm. Read `threat_name` and `action`. A block or reset means it was stopped. An alert or allow means it got through.

Respond. If it got through, hunt on the internal host with the Splunk pivots (`dest_ip` for inbound, `src_ip` for outbound). For command-and-control going out to an internet IP, block that destination: it isn't the scorer.

Health and Splunk

Whether you can still see everything, and whether Splunk itself, a scored service, is safe. Dashboard: Health & Splunk (`/app/ccdc_hunts/health`). Its Searches by user panel shows exactly what the Splunk Query check looks for: never delete, rename or filter out that data.

Host stopped sending logs

Severity 5 · Splunk's own data

A host on the expected-hosts list sent nothing for 10 minutes, or never has. The box is down, its forwarder or Sysmon was stopped, or the path to Splunk broke. Every other alert is blind for that host until it's fixed.

Confirm. `minutes_quiet` says how long. "never seen" usually means a typo in the list, or a host that isn't set up yet. Edit the list in Settings → Advanced search → Search macros → `expected_hosts`.

Respond.

- Windows: `Get-Service SplunkForwarder, Sysmon64`, then `Start-Service` whichever is stopped.
- Linux: `sudo systemctl status SplunkForwarder`, then `sudo systemctl start SplunkForwarder`
- Palo Alto or VyOS: check nobody changed its syslog settings.
- When logs come back, the forwarder sends the backlog. Hunt the quiet window on that host: it's when they did something.

Splunk login failures

Severity 4 · Splunk audit log

Someone failed to log into Splunk. Splunk admin can run commands on the Debian box through apps, so red teams go after it.

Confirm. `user` and `src` show who and from where. The scorer only fails if its Splunk password changed.

Respond.

1. If the Splunk `admin` password is still the default or the team-packet one, change it: Settings → Users → admin → Edit. Change the scoring account only through the competition's process.
2. On the Health dashboard, check Splunk users (new admins?) and Installed apps (new apps?). Disable a rogue app in Apps → Manage Apps.
3. Leave port 8089 open, because it's scored. Strong passwords are the protection.

Persistence sweeps

Run the matching list on the host after containment. The red team usually leaves a way back in before you notice them. Tick each box when it comes back clean.

Windows (admin PowerShell)

- [ ] **Services** running a shell, a script, or something from Temp or ProgramData: `Get-CimInstance Win32_Service | Where-Object PathName -match 'cmd|powershell|wscript|\.bat|Temp|ProgramData' | Select-Object Name, StartMode, PathName`
- [ ] **Scheduled tasks** in the root folder or running a script host: `Get-ScheduledTask | Where-Object { $_.TaskPath -eq '\' -or $_.Actions.Execute -match 'cmd|powershell|wscript|cscript|mshta' } | Select-Object TaskPath, TaskName, State`
- [ ] **Hidden tasks** (in the registry, invisible in Task Scheduler): `Get-ChildItem 'HKLM:\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tree' -Recurse | Where-Object { (Get-ItemProperty $_.PSPath -Name Id -EA 0) -and -not (Get-ItemProperty $_.PSPath -Name SD -EA 0) } | Select-Object Name`
- [ ] **Run keys**: `Get-ItemProperty 'HKLM:\Software\Microsoft\Windows\CurrentVersion\Run','HKLM:\Software\Microsoft\Windows\CurrentVersion\RunOnce'`

- [ ] **Startup folder**: `Get-ChildItem "$env:ProgramData\Microsoft\Windows\Start Menu\Programs\StartUp"`
- [ ] **WMI subscriptions** ("SCM Event Log Consumer" is normal): `Get-CimInstance -Namespace root\subscription -ClassName __EventConsumer | Select-Object Name, CommandLineTemplate`
- [ ] **Admin groups**: `Get-ADGroupMember "Domain Admins" -Recursive | Select-Object SamAccountName` (member servers: `Get-LocalGroupMember Administrators`)
- [ ] **Sticky-keys backdoor** (any output is bad): `'sethc.exe','utilman.exe','osk.exe','narrator.exe','magnify.exe','display switch.exe' | ForEach-Object { Get-ItemProperty "HKLM:\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options\$_" -Name Debugger -EA 0 }`
- [ ] **Hiding spots** the invitational red team used, files from the last 6 hours: `Get-ChildItem "$env:ProgramData\Microsoft\Windows\WER","$env:ProgramData\Microsoft\Diagnosis","$env:ProgramData\Microsoft\Windows\Power Efficiency Diagnostics","$env:windir\Temp" -Recurse -File -Force -EA 0 | Where-Object LastWriteTime -gt (Get-Date).AddHours(-6) | Select-Object FullName, LastWriteTime`
- [ ] **BITS jobs**: `bitsadmin /list /allusers /verbose`
- [ ] **Defender** on, no planted exclusions: `Get-MpPreference | Select-Object ExclusionPath, ExclusionProcess, DisableRealtimeMonitoring`
- [ ] **Logging** running: `Get-Service Sysmon64, SplunkForwarder | Select-Object Name, Status`

On the MSSQL box, also check SQL Agent jobs, startup procedures, server triggers and the sysadmin role.

Linux (root shell)

- [ ] **Extra root accounts** (only `root` should print): `awk -F: '$3 == 0' /etc/passwd`
- [ ] **Accounts that can log in**: `grep -vE 'nologin|false' /etc/passwd`
- [ ] **Admins and sudo rules**: `getent group sudo wheel; cat /etc/sudoers; ls -la /etc/sudoers.d`
- [ ] **SSH keys** (read every file it lists): `find / -name authorized_keys -not -path '/proc/*' 2>/dev/null`

- [ ] **Cron**: `cat /etc/crontab; ls -la /etc/cron.d /var/spool/cron /var/spool/cron/crontabs 2>/dev/null`
- [ ] **New services and timers** (last 2 days): `find /etc/systemd/system /usr/lib/systemd/system -mtime -2 \( -name '*.service' -o -name '*.timer' \)`
- [ ] **Listening ports** and the program behind each: `ss -tulpn`
- [ ] **Open outbound connections** (beacons): `ss -tunp state established`
- [ ] **Shell startup files** with download or reverse-shell code: `grep -rnE 'curl|wget|nc|bash -i|/dev/tcp' /etc/profile /etc/profile.d /etc/bash.bashrc /root/.bashrc /home/*/.bashrc 2>/dev/null`
- [ ] **Preload rootkit** (the file shouldn't exist): `cat /etc/ld.so.preload`
- [ ] **Web root** files changed in 2 hours: `find /var/www -type f -mmin -120`
- [ ] **Logging** running: `systemctl status SplunkForwarder rsyslog auditd`

Splunk pivots

Start from the hit's host and time, then pivot to the account and source IP. To set the time fast, click the hit's timestamp in the Events tab and pick a "Nearby Events" window.

The app's macros name where each log lives: `windows_security`, `windows_system`, `windows_sysmon`, `windows_powershell`, `windows_defender`, `windows_winrm`, `linux_auth`, `linux_syslog`, `web_access`, `mail`, `ftp`, `paloalto_traffic`, `paloalto_threat`, `paloalto_system`, `paloalto_config`, `vyos` and `splunk_audit`. Type them inside backticks. If a log moves to a different index, change it once in Settings → Advanced search → Search macros.

Everything on the host around the hit:

Run in Splunk »

```
index=* host="<host>" | table _time source EventCode User Image CommandLine _raw
```

Every program started on a Windows host, and what started it:

Run in Splunk »

```
`windows_sysmon` host="<host>" EventCode=1 | table _time User ParentImage Image CommandLine
```

Every login and sudo on a Linux host:

Run in Splunk »

```
`linux_auth` host="<host>" ("Accepted" OR "Failed password" OR "COMMAND=") |
table _time _raw
```

Everything that mentions the source IP, across all hosts (a plain keyword search, so it works whatever the field is called):

Run in Splunk »

```
index=* "<source IP>" | stats count min(_time) as first max(_time) as last by
host source | convert ctime(first) ctime(last)
```

Everything that mentions the account:

Run in Splunk »

```
index=* "<account>" | stats count by host source EventCode
```

Where a Windows host connected out to:

Run in Splunk »

```
`windows_sysmon` host="<host>" EventCode=3 | stats count by Image
DestinationIp DestinationPort | sort - count
```

Incident report

Injects often ask for an incident report by a deadline. Fill in the Entry column as you work, not afterward, and copy the table for each new incident.

Field	What to write	Entry
Detected	Time of the first alert, and which alert fired	
Host(s)	Every box involved	
Source IP	Where the logons came from	
Account used	The account in the logon at the same minute	
What the attacker did	Their commands, copied from the alert's command column	
Impact	Accounts created, files dropped, data touched, scored services affected	
Containment	Passwords reset and sessions closed, with times	
Eradication	Services, tasks, keys and files disabled or quarantined, with hashes	

Field	What to write	Entry
Status	Contained, or still monitoring	
Evidence	Screenshots of the Splunk results, quarantine folder path	

Sources

- NetExec smbexec source: the service, `.bat` and output-file names the remote execution alerts look for
- NetExec wmiexec source and atexec source: NetExec's other two routes